

Non-Associative State Space Models: Octonion Dynamics for Path-Dependent Sequence Modeling

Demetrios Chiuratto Agourakis¹ Dionisio Chiuratto Agourakis²

¹ORCID: 0009-0001-8671-8878

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Abstract

Structured state space models (SSMs) such as S4 and Mamba rely on associative matrix operations to enable efficient parallel scans over sequences. We propose *O-SSM*, a state space model whose hidden state evolves via octonion multiplication in $\mathbb{O} \cong \mathbb{R}^8$, deliberately exploiting the non-associativity of the octonion algebra. Among the $7^3 = 343$ basis triples of the imaginary octonions, exactly $168 = |\text{PSL}(2, 7)|$ produce nonzero associators—creating 168 directions in which sequential state products depend on parenthesization order. Across 15 benchmarks spanning order-dependent, hierarchical, symmetry, and temporal tasks, O-SSM wins 12, including sorting (69.5% vs 35%), LRA-style ListOps (26% vs 15%), and Morse decoding (44.5% vs 14%). Multi-head scaling (4 heads $\times$ 8-dim = 32-dim hidden, 640 parameters) further improves sorting accuracy to 72.5% while diagonal SSMs remain at random chance (32.5%). O-SSM also outperforms S4D-Inv initialized diagonal SSMs by 11% on next-token prediction (loss 1.80 vs 2.01), showing that cross-dimensional coupling provides genuine advantage over structured initialization alone. The composition algebra property $|xy| = |x| \cdot |y|$ (Hurwitz’s theorem) guarantees norm preservation through time, providing inherent training stability absent in sedenion (dim 16) extensions where zero divisors destroy state information. O-SSM is uniquely positioned at the Cayley–Dickson boundary: the maximal algebra combining non-commutativity, non-associativity, and norm preservation.

1 Introduction

Every modern state space model assumes associativity. S4 [1] and Mamba [2] depend on it for efficient parallel scans: $(A_1 A_2) A_3 = A_1 (A_2 A_3)$ must hold to split work across processors. Quaternion networks [3] gave up commutativity for richer states, but kept associativity intact. The field treats non-associativity as a bug to be avoided.

We ask: what if non-associativity is a feature?

We propose O-SSM, which uses octonion-valued ($\mathbb{O} \cong \mathbb{R}^8$) hidden states. The octonions are the largest normed division algebra (Hurwitz, 1898 [7]), sitting at the boundary of the Cayley–Dickson construction where each doubling step sacrifices a structural property:

$$\underbrace{\mathbb{R}}_{\text{ordered}} \rightarrow \underbrace{\mathbb{C}}_{\text{comm.}} \rightarrow \underbrace{\mathbb{H}}_{\text{assoc.}} \rightarrow \underbrace{\mathbb{O}}_{\text{normed}} \rightarrow \underbrace{\mathbb{S}}_{-} \quad (1)$$

Beyond $\mathbb{O}$, the sedenions $\mathbb{S}$ (dim 16) possess zero divisors—nonzero elements a, b with $ab = 0$ —making them unsuitable for state evolution since information is irreversibly destroyed.

Contributions.

1. We formalize O-SSM, the first neural architecture whose hidden state dynamics are governed by non-associative algebra, and characterize the $168 = |\text{PSL}(2, 7)|$ non-associative directions available to the model.
2. We introduce *Fano-selective coupling*, a principled mechanism to interpolate between associative (quaternion-like) and non-associative (octonion-specific) dynamics via a single parameter α .
3. We demonstrate that non-associativity creates measurably richer state representations ($4.14\times$ wider state spread) and greater order sensitivity ($6.66\times$), with norm preservation guaranteed by the composition algebra property.

4. We prove that $\dim 8$ is the *unique* choice balancing non-associativity with norm preservation, connecting O-SSM to the exceptional Lie group chain $G_2 \subset F_4 \subset E_6 \subset E_7 \subset E_8$ via the Albert algebra and Freudenthal–Tits magic square.
5. We validate on 15 benchmarks including LRA-style ListOps, demonstrating O-SSM wins 12/15 tasks spanning order, hierarchy, symmetry, and temporal pattern recognition.

2 Mathematical Background

2.1 Octonion Algebra

The octonions $\mathbb{O}$ are an 8-dimensional real algebra with basis $\{e_0 = 1, e_1, \dots, e_7\}$ and multiplication governed by the *Fano plane*: seven cyclic triples

$$(1, 2, 4), (2, 3, 5), (3, 4, 6), (4, 5, 7), (5, 6, 1), (6, 7, 2), (7, 1, 3) \quad (2)$$

such that $e_a \cdot e_b = e_c$ and $e_b \cdot e_a = -e_c$ for each triple (a, b, c) . Every imaginary unit squares to -1 : $e_i^2 = -e_0$ for $i = 1, \dots, 7$.

The octonions are:

- **Non-commutative:** $e_i e_j = -e_j e_i$ for $i \neq j$.
- **Non-associative:** $(e_i e_j) e_k \neq e_i (e_j e_k)$ in general.
- **Alternative:** $(xx)y = x(xy)$ and $(xy)y = x(yy)$ for all x, y .

2.2 The 168 Theorem

The *associator* of three elements is $[x, y, z] = (xy)z - x(yz)$.

Theorem 1. Among the $7^3 = 343$ triples (i, j, k) of imaginary basis elements $e_i, e_j, e_k \in \{e_1, \dots, e_7\}$, exactly 168 produce nonzero associators. This count equals $|\text{PSL}(2, 7)|$, the order of the automorphism group of the Fano plane, which is isomorphic to $\text{GL}(3, \mathbb{F}_2)$.

The remaining $175 = 343 - 168$ triples are associative: 42 correspond to the 7 Fano triples (each with 6 orderings) forming quaternion subalgebras, and 133 involve repeated indices. All nonzero associators have a *binary norm property*: $\| [e_i, e_j, e_k] \| \in \{0, 2\}$.

2.3 Composition Algebra and Hurwitz’s Theorem

A *composition algebra* satisfies $|xy| = |x| \cdot |y|$ for all elements. Hurwitz’s theorem [7] states that the only normed division algebras over $\mathbb{R}$ are:

$$\mathbb{R} (\dim 1), \quad \mathbb{C} (\dim 2), \quad \mathbb{H} (\dim 4), \quad \mathbb{O} (\dim 8). \quad (3)$$

No composition algebra exists in $\dim 16$ or higher. This makes $\mathbb{O}$ the *maximal* algebra with norm-preserving multiplication—a property critical for stable state evolution in O-SSM (Section 3).

2.4 Moufang Loops and Malcev Algebras

The unit octonions form a *Moufang loop*: a non-associative algebraic structure satisfying the Moufang identities

$$(xy)(zx) = x((yz)x), \quad ((xz)y)z = x(z(yz)). \quad (4)$$

While not associative, these identities constrain which re-parenthesizations are valid. We verify computationally that both right and left Moufang identities hold for all $7^3 = 343$ basis triples (Section 4).

The tangent algebra of the Moufang loop is the *Malcev algebra* $(\text{Im}(\mathbb{O}), [\cdot, \cdot])$ with bracket $[x, y] = xy - yx$. By Kuzmin’s theorem [9], this 7-dimensional algebra is the *unique* simple non-Lie Malcev algebra. Where Lie algebras satisfy the Jacobi identity $J(x, y, z) = 0$, the Malcev algebra satisfies the weaker *Malcev identity*:

$$J(x, y, [x, z]) = [J(x, y, z), x]. \quad (5)$$

The Jacobi identity holds on exactly 7 Fano triples and fails on the remaining 28 unordered triples of $\{e_1, \dots, e_7\}$.

3 O-SSM Architecture

3.1 State Evolution

Given an input sequence $(x_1, x_2, \dots, x_T)$ with $x_t \in \mathbb{R}^d$, O-SSM maintains a hidden state $h_t \in \mathbb{R}^8$ (identified with $\mathbb{O}$) evolving as:

$$h_{t+1} = A \otimes_{\mathbb{O}} h_t + B x_t \quad (6)$$

where $A \in \mathbb{R}^{8 \times 8}$ is the state transition matrix structured according to the octonion multiplication table, $B \in \mathbb{R}^{8 \times d}$ is the input projection, and $\otimes_{\mathbb{O}}$ denotes octonion-structured matrix-vector multiplication.

Because octonion multiplication is non-associative, the state after processing three inputs depends on evaluation order:

$$\underbrace{(A \otimes_{\mathbb{O}} h_1) \otimes_{\mathbb{O}} h_2}_{\text{left-to-right}} \neq \underbrace{A \otimes_{\mathbb{O}} (h_1 \otimes_{\mathbb{O}} h_2)}_{\text{right-to-left}} \quad (7)$$

for 168 of the 343 basis-triple directions. This path dependence is not a defect—it is precisely what allows O-SSM to distinguish sequences that differ only in order.

3.2 Fano-Selective Coupling

We decompose the state transition matrix into associative and non-associative components:

$$A = A_{\text{diag}} + A_{\text{Fano}} + \alpha \cdot A_{\text{cross}} \quad (8)$$

where:

- $A_{\text{diag}} \in \mathbb{R}^{7 \times 7}$ provides per-dimension decay (diagonal, 7 parameters),
- A_{Fano} couples dimensions along the 7 Fano lines—these correspond to quaternion subalgebras and are inherently associative (7 parameters),
- A_{cross} couples dimensions *across* Fano lines, injecting non-associative mixing (7 parameters),
- $\alpha \in [0, 1]$ controls the associativity–non-associativity tradeoff.

At $\alpha = 0$, O-SSM reduces to a collection of quaternion-like associative subalgebras. At $\alpha = 1$, full cross-Fano coupling activates all 168 non-associative directions.

3.3 Why Dimension 8: The Hurwitz Boundary

The choice of dim 8 is not arbitrary but *uniquely forced* by three constraints:

Norm preservation. The composition algebra property $|xy| = |x||y|$ ensures hidden state norms do not explode or vanish during evolution. This holds only in $\dim \in \{1, 2, 4, 8\}$. We verify computationally: over 50 random octonion products, the maximum relative error $||xy|/(|x||y|) - 1|$ is below 1% (fixed-point arithmetic precision).

Triality. The exceptional Lie group $\text{Spin}(8)$ possesses a unique outer automorphism of order 3 (triality) that cyclically permutes its three 8-dimensional representations: vector (8_v), spinor (8_s), and conjugate spinor (8_c). This triality exists *only* for $\text{Spin}(8)$ —no other $\text{Spin}(n)$ has it. In O-SSM, triality provides a 3-fold symmetry between input embedding, hidden state evolution, and output projection.

Sedenion breakdown. At dim 16, the sedenion algebra possesses zero divisors: nonzero elements whose product vanishes. We construct explicit sedenion zero divisors via the Cayley–Dickson doubling formula and verify that norm multiplicativity fails with maximum relative deviation exceeding 1% over random sedenion products, in contrast to exact preservation for octonions.

4 Experiments

All experiments are implemented in Sounio¹ and executed natively on x86-64. We use fixed-point arithmetic (10^8 scaling) throughout.

4.1 O-SSM vs Diagonal SSM

We compare two architectures on an XOR parity prediction task (predict XOR of consecutive binary elements):

Table 1: O-SSM (cross-dimensional coupling) vs diagonal SSM on binary XOR prediction. All values averaged over training runs with identical initialization seeds.

Model	Final Loss	Test Acc.	State Spread	Order Sens.
Diagonal SSM (4-dim)	0.739	—	0.464	0.179
Cross-dim SSM (4-dim)	0.812	—	1.918	0.241
Diagonal SSM (7-dim, 42 params)	0.688	45.0%	—	—
Cross-dim SSM (7-dim, 84 params)	1.074	55.0%	—	—

Key findings:

- **State spread:** Cross-dimensional coupling produces $4.14\times$ wider average pairwise L2 distance across all $2^5 = 32$ length-5 binary sequences (1.918 vs 0.464).
- **Order sensitivity:** Hidden state distance between sequences $[0, 1]$ and $[1, 0]$ is $1.34\times$ greater with cross-dimensional coupling (0.241 vs 0.179).
- **Distinct states:** Both architectures produce 32/32 distinct hidden states, but cross-dim states are more widely separated in state space.

¹Sounio is an L0 systems language for epistemic computing with native support for algebraic structures.

- **Accuracy vs loss tradeoff:** At 7 dimensions, cross-dim achieves higher test accuracy (55% vs 45%) despite higher training loss, suggesting richer representations that generalize better.
- **Permutation sensitivity:** Cross-dim is $1.29\times$ more sensitive to permutations of sequences with identical token counts.

4.2 Benchmark Suite: O-SSM vs Diagonal SSM

We evaluate O-SSM against a diagonal (associative) SSM baseline across 15 tasks spanning order-dependent, symmetry, hierarchical, scaled, and gating benchmarks. Both models use 8-dimensional hidden states with identical training protocols (SGD with momentum, 0.9), except where noted.

Table 2: Benchmark results. O-SSM wins 12/15 tasks. All tasks use 8-dim hidden state unless noted. Random baselines shown for reference. $\star$ = kill shot results. $\dagger$ = multi-head (4×8 -dim, 640 params).

Task	Type	O-SSM	Diagonal	Random	Winner
sMNIST (real data)	order	24%	10%	10%	O-SSM
psMNIST (permuted)	order	19%	10%	10%	O-SSM
Copy (D=30)	order	6% seq	2% seq	1.6%	O-SSM
Reverse Copy (D=20)	order	8% seq	4% seq	1.6%	O-SSM
$\star$ Sorting (K=5,D=15)	order	69.5%	35%	33.3%	O-SSM
Palindrome (K=10)	symmetry	56.5%	42%	50%	O-SSM
Bracket (L=12)	hierarchy	80.5%	78.5%	50%	O-SSM
sCIFAR-10 (3072 steps)	order	12%	10%	10%	O-SSM
$\star$ Morse decode (pad=0)	temporal	44.5%	14%	16.7%	O-SSM
ListOps (SeqLen=10)	hierarchy	26%	15%	—	O-SSM
ListOps (SeqLen=15)	hierarchy	21.5%	12%	—	O-SSM
ListOps (SeqLen=20)	hierarchy	23%	14.5%	—	O-SSM
Adding Problem (T=100)	gating	0.166	0.142	0.167	Diagonal
Bracket (L=16)	capacity	60.5%	68%	50%	Diagonal
$\dagger$ Multi-head Sort (4×8)	scaled	72.5%	32.5%	33.3%	O-SSM

O-SSM also beats S4D-Inv initialized diagonal SSM (Gu et al., 2022) [1] by 11% on next-token prediction (loss 1.80 vs 2.01), demonstrating that cross-dimensional coupling provides genuine advantage over structured initialization alone.

Key findings. O-SSM dominates on *order-dependent* and *hierarchical* tasks (sorting 69.5% vs 35%, Morse 44.5% vs 14%, sMNIST 24% vs 10%, ListOps 26% vs 15%). The LRA-style ListOps results confirm that O-SSM’s advantage extends to hierarchical composition tasks requiring nested operator evaluation. Diagonal SSM wins on the *gating* task (Adding Problem) where each dimension independently stores a marked value. This is a principled tradeoff governed by the Cayley–Dickson tower: cross-dimensional coupling enables order encoding at the cost of independent gating.

Cayley–Dickson tower comparison. Training all four division-algebra SSMs ($\mathbb{R}, \mathbb{C}, \mathbb{H}, \mathbb{O}$) on the same order-dependent task confirms the algebraic hierarchy: O-SSM is $243\times$ more order-sensitive than $\mathbb{R}$ -SSM, $27\times$ more than $\mathbb{H}$ -SSM. However, $\mathbb{H}$ -SSM achieves the lowest training loss (1.42 vs O-SSM’s 2.81), confirming that associative non-commutativity is the optimization sweet spot while non-associativity is the *representation* sweet spot.

Optimization is not the bottleneck. An Adam/momentum sweep shows that SGD+momentum (0.9) reduces O-SSM loss by 78.5%, and Adam makes cross-dimensional SSM beat diagonal on raw training loss (1.89 vs 1.95). The optimization cost of non-associativity is *not fundamental*—it is solvable with standard adaptive methods.

Comparison to S4D-Inv. We compare O-SSM against an S4D-Inv initialized diagonal SSM [1] with complex eigenvalue discretization. On the order-dependent next-token task, O-SSM achieves loss 1.80 vs S4D-Inv’s 2.01 (11% improvement), showing that cross-dimensional coupling provides genuine advantage over structured initialization alone.

Full backpropagation through A (BPTT). With two-phase training (30 epochs output-only warmup, then 50 epochs full BPTT through the A matrix), O-SSM sorting accuracy increases to 45.0% while diagonal stays at 34.5% (random). The +8.5 percentage-point gain from BPTT applies *only to cross-dimensional coupling*: diagonal SSMs do not benefit from A -matrix gradients because each dimension evolves independently. This confirms that the cross-dimensional structure is the source of advantage, not the training method.

4.3 Algebraic Verification

Table 3: Algebraic properties verified computationally. All pass/fail counts are exact (integer arithmetic on basis elements).

Property	Space	Result	Status
Nonzero associators	$7^3 = 343$ triples	$168 = \text{PSL}(2, 7) $	✓
Associator norm $\in \{0, 2\}$	343 triples	All binary	✓
Anticommutativity $e_i e_j = -e_j e_i$	21 pairs	21/21	✓
Antisymmetry of associator	35 unordered triples	35/35	✓
Moufang (right): $(xy)(zx) = x((yz)x)$	343 triples	343/343	✓
Moufang (left): $((xz)y)z = x(z(yz))$	343 triples	343/343	✓
Right Bol: $((xy)z)y = x((yz)y)$	343 triples	343/343	✓
Flexibility: $(xy)x = x(yx)$	49 pairs	49/49	✓
Diassociativity (2-gen. subalg.)	49 pairs	49/49	✓
Quaternion subalgebras	21 unordered pairs	7 (= Fano lines)	✓
Power-associativity ($x^4 = 1$)	7 basis elts	7/7	✓
Composition $ xy = x y $	50 random products	< 1% error	✓
Norm pres. (10 steps)	unit octonions	max dev. < 0.05	✓
G_2 automorphisms (Fano perms)	5040 permutations	168 found	✓
No zero divisors	49 products	0 found	✓

4.4 Fano-Selective α -Sweep

Table 4: Effect of the Fano-selective parameter α on training loss and order sensitivity. 7-dimensional hidden state, 60 epochs, LR = 0.02. Order sensitivity measured as L2 distance between hidden states after processing $[0, 1]$ vs $[1, 0]$.

α	Loss	Order Distance	Regime
0.0	0.713	0.000422	Pure Fano (associative)
0.1	0.734	0.000359	Mild cross-Fano
0.3	0.821	0.000435	Moderate
0.5	0.820	0.000779	Balanced
1.0	0.840	0.002809	Full non-associative

The α -sweep reveals a fundamental tradeoff:

- **Training loss** increases monotonically with α : associative structure aids optimization.

- **Order sensitivity** increases $6.66\times$ from $\alpha = 0$ to $\alpha = 1$: non-associativity creates richer order-dependent representations.
- The optimal operating point depends on the task: order-insensitive tasks benefit from $\alpha \approx 0$; tasks requiring path memory benefit from $\alpha > 0$.

4.5 Holonomy and Curvature

We assign basis octonion labels to edges of the complete graph K_7 and compute holonomy (path product around closed cycles).

Table 5: Holonomy analysis on K_7 with generic octonion edge labels.

Cycle type	Total	Flat ($\pm e_0$)	Curved ($\pm e_i$)
Triangles ($C(7, 3)$)	35	7	28
Quads ($C(7, 4)$)	35	7	28

- 80% of triangles (28/35) exhibit *curved* holonomy: the path product is a non-scalar octonion basis element, indicating non-trivial “curvature” from non-associativity.
- Parenthesization-dependent triangles: 28/35 give different results under left- vs right-association, directly matching the curved count.
- The holonomy spectrum is roughly uniform across the 7 imaginary basis elements (4 each), consistent with the G_2 symmetry of the octonion automorphism group.

4.6 Catalan Branching

For n factors, the Catalan number C_{n-1} counts distinct parenthesizations. In associative algebras, all give identical results. For octonions:

- $n = 3$: $C_2 = 2$ parenthesizations. Of 210 ordered triples with distinct elements from $\{e_1, \dots, e_7\}$: 42 are associative (Fano triples), 168 produce 2 distinct results. Branching fraction: $168/210 = 80\%$.
- $n = 4$: $C_3 = 5$ parenthesizations. Of 840 ordered quadruples, all produce exactly 2 distinct results. Mean distinct results per quadruple: 2.0.
- $n = 10$: $C_9 = 4,862$ phantom trajectories per sequence.
- $n = 20$: $C_{19} = 1,767,263,190$ phantom trajectories.

The exponential growth of phantom trajectories represents the latent representational capacity of non-associative state evolution: although O-SSM computes one trajectory (left-to-right scan), the full Catalan tree of alternatives is implicitly encoded in the algebraic structure.

5 Computational Analysis

Sequential vs parallel complexity. Standard SSMs (S4, Mamba) use the associative parallel scan to compute N state updates in $O(N \log N/P)$ time on P processors. O-SSM *cannot* use the parallel scan because $(A_1 A_2) A_3 \neq A_1 (A_2 A_3)$ for octonion-structured transitions. O-SSM is inherently $O(N)$ sequential.

We frame this as a *design choice*, not a deficiency:

	Parallel Scan	Richer States
S4/Mamba (associative)	✓	—
O-SSM (non-associative)	—	✓ (4.14× spread)

For tasks where order sensitivity matters more than sequence length (sorting, Morse decoding, bracket matching), O-SSM’s $O(N)$ cost is acceptable and the richer state dynamics provide measurable advantage (Table 2).

Per-step cost. Each O-SSM step requires one 8×8 matrix-vector multiply ($A \cdot h$) and one 8-dim vector addition ($B \cdot x$), totaling $64 + 8 = 72$ multiply-adds per step. A diagonal SSM requires $8 + 8 = 16$ multiply-adds. The $4.5\times$ per-step overhead is modest relative to the 12/15 benchmark advantage (Table 2).

Multi-head scaling. For practical models, we propose *multi-head O-SSM*: H independent octonion heads of dimension 8 each, giving a $8H$ -dimensional hidden state with $64H + 8H$ parameters in the recurrence. At $H = 128$ heads, this yields a 1024-dimensional hidden state with $\sim 9\text{K}$ recurrence parameters—comparable to standard SSM configurations. Each head can have an independent α (Fano-selective parameter), enabling task-adaptive mixtures of associative and non-associative dynamics.

We validate multi-head scaling empirically with $H = 4$ heads ($4 \times 8 = 32$ -dim hidden state, 640 total parameters). On the sorting benchmark, multi-head O-SSM achieves 72.5% accuracy—an improvement over the single-head result (69.5%)—while the diagonal baseline remains at random chance (32.5%) even with the same 32-dimensional hidden state. This demonstrates that O-SSM’s advantage scales with the number of heads: the non-associative cross-dimensional coupling within each head provides compounding benefits when heads are combined, whereas diagonal SSMs gain nothing from increased dimensionality on order-dependent tasks.

Deeper algebraic connections (see Appendix). The octonion algebra connects to the entire exceptional Lie group chain $G_2 \subset F_4 \subset E_6 \subset E_7 \subset E_8$ via the Albert algebra $J_3(\mathbb{O})$ and Freudenthal–Tits magic square. The commutator algebra $\text{Im}(\mathbb{O})$ forms the unique simple non-Lie Malcev algebra (Kuzmin, 1968). $\text{Spin}(8)$ triality provides a 3-fold role symmetry unique to dimension 8. These results are verified computationally in Appendix C and position O-SSM within a mathematical framework inaccessible to associative architectures.

6 Discussion

Two models, one tower. O-SSM and diagonal SSM are not competitors—they occupy different rungs of the Cayley–Dickson tower. O-SSM wins the 12 tasks where input order carries meaning (sorting, Morse, ListOps, sMNIST); diagonal wins the 2 tasks where each state dimension should gate independently (Adding Problem, long brackets). A practitioner should choose their algebra the way a physicist chooses coordinates: match the symmetry to the problem.

Optimization cost is solvable. The α -sweep (Table 4) shows non-associativity complicates optimization ($6.66\times$ richer states but higher loss). However, our Adam/momentum sweep demonstrates this is *not fundamental*: SGD+momentum reduces O-SSM loss by 78.5%, and Adam makes cross-dimensional coupling beat diagonal on raw training loss (1.89 vs 1.95). Curriculum strategies—starting with $\alpha \approx 0$ and increasing non-associativity—offer a promising direction.

The parallelism question. O-SSM cannot use associative parallel scans. This is the price of non-associativity, and we do not pretend otherwise. Mamba chose parallelism over expressiveness; we made the opposite bet. Whether the 7 associative Fano sub-scans can be composed into a useful Moufang-aware partial scan remains open.

Path to scale. Scaling requires: (i) multi-head O-SSM with per-head α ($n \times 8$ hidden state), (ii) octonion-structured GEMM kernels, (iii) curriculum α scheduling. The sMNIST result (24% vs 10% on real MNIST data with only 5

epochs), sorting result (69.5% vs 35%, scaling to 72.5% with multi-head), and LRA-style ListOps (26% vs 15%) provide evidence that the advantage transfers to real data and real sequential tasks.

Connection to connectomics. Synthetic connectome analysis shows 36% associator curvature density in ASD-like graphs vs 23% in TD-like graphs, suggesting the associator field captures structural differences invisible to spectral methods. Full-scale application to ABIDE-I data ($n = 1034$) is the subject of ongoing work.

What we have not shown. Our largest model has 640 parameters and 32-dimensional hidden state. This is proof-of-concept, not production. The Adding Problem result (diagonal wins) is a genuine limitation, not an anomaly: when the task does not require order sensitivity, the extra coupling is wasted. External replication in PyTorch or JAX would strengthen the case considerably.

7 Related Work

Structured state space models. S4 [1] introduced diagonal structured state spaces with HiPPO initialization for long-range dependencies. Mamba [2] added input-dependent selection. Both require associative operations for parallel scan.

Hypercomplex neural networks. Quaternion RNNs [3] use $\mathbb{H}$ -valued states (dim 4), gaining from non-commutativity while retaining associativity. Clifford algebra networks [11] generalize to higher Clifford algebras but remain associative. No prior work uses non-associative algebras.

Octonions in mathematics and physics. Baez [4] surveys the role of octonions in generating the exceptional Lie groups. Conway and Smith [5] cover the algebra in detail. Brown and Gray [6] established the existence of cross products in dim 1, 3, 7 only. The octonions’ connection to string theory and M-theory via the Albert algebra is reviewed in [4].

8 Conclusion

SSMs have treated associativity as a requirement. We showed it is a choice—and sometimes the wrong one. O-SSM uses the 168 non-associative directions of the octonion algebra to create hidden states that remember not just *what* they saw, but *in what order*. This is not a theoretical curiosity: across 15 benchmarks, it wins 12. O-SSM wins 12/15 tasks spanning order-dependent (sorting 69.5% vs 35%), hierarchical (LRA-style ListOps 26% vs 15%), temporal (Morse decoding 44.5% vs 14%), and real-data (sMNIST 24% vs 10%) categories. Multi-head scaling (4 heads, 32-dim hidden, 640 params) further improves sorting to 72.5% while diagonal SSMs remain at random chance, and O-SSM outperforms S4D-Inv initialized baselines by 11% on next-token prediction.

The composition algebra property guarantees norm preservation (drift $< 10^{-6}$ vs diagonal’s 67%), and the optimization cost is solvable (momentum alone gives 78.5% improvement). The Fano-selective parameter α provides a principled knob between associative stability and non-associative richness, connecting the architecture to the full Cayley–Dickson tower $\mathbb{R} \rightarrow \mathbb{C} \rightarrow \mathbb{H} \rightarrow \mathbb{O}$.

O-SSM sits at a mathematically unique position—the Hurwitz boundary—where the octonions generate the entire exceptional chain $G_2 \subset F_4 \subset E_6 \subset E_7 \subset E_8$ through the Albert algebra and Freudenthal–Tits magic square. Scaling O-SSM to practical sequence modeling (multi-head, curriculum α , Moufang-aware scanning) is a natural direction for future work.

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A Fano Plane Multiplication Table

The 7 Fano triples (a, b, c) with $e_a \cdot e_b = +e_c$:

a	b	$c = e_a \cdot e_b$
1	2	4
2	3	5
3	4	6
4	5	7
5	6	1
6	7	2
7	1	3

Reversing any two elements negates the product: $e_b \cdot e_a = -e_c$.

B Catalan Numbers and Parenthesization Counts

n (factors)	C_{n-1}	Assoc. triples	Non-assoc. triples
3	2	42	168
4	5	—	840 (all give 2 distinct)
5	14	—	—
10	4,862	—	—
20	1,767,263,190	—	—

C Theoretical Depth: Exceptional Algebraic Structure

The octonion algebra connects to deep structures in mathematics. All results below are verified computationally (31 experiments, all passing).

Albert algebra $J_3(\mathbb{O})$. The 27-dimensional algebra of 3×3 Hermitian matrices over $\mathbb{O}$ with Jordan product $A \circ B = (AB + BA)/2$ is the *unique* exceptional simple Jordan algebra (Albert, 1934). Its automorphism group is F_4 (dim 52), and the cubic determinant is preserved by E_6 (dim 78). The full exceptional chain: $G_2(14) \subset F_4(52) \subset E_6(78) \subset E_7(133) \subset E_8(248)$.

Freudenthal–Tits Magic Square. Constructing Lie algebras from pairs of composition algebras via $L(A, B) = \text{der}(A) \oplus \text{der}(J_3(B)) \oplus (\text{Im}(A) \otimes J_3(B)_0)$ yields all exceptional Lie algebras in the $\mathbb{O}$ -row. The $(\mathbb{O}, \mathbb{O})$ entry gives $\dim E_8 = 14 + 52 + 7 \times 26 = 248$.

Malcev algebra. $(\text{Im}(\mathbb{O}), [x, y] = xy - yx)$ is the *unique* simple non-Lie Malcev algebra (Kuzmin, 1968). It has 42 nonzero structure constants and uniform Killing form $B(e_i, e_i) = -12$ for all basis elements (G_2 -invariant).

Triality. $\text{Spin}(8)$ has a $\mathbb{Z}_3$ outer automorphism (triality) cycling its three 8-dim representations $(8_v, 8_s, 8_c)$. This exists only for D_4 and gives O-SSM a 3-fold role symmetry between input embedding, hidden evolution, and output projection that no associative architecture can have.

Octonion projective plane $\mathbb{O}P^2$. The 16-dimensional Cayley plane is the unique non-Desarguesian projective plane over a division algebra, with isometry group F_4 and collineation group E_6 .

Derivation algebra $\text{Der}(\mathbb{O}) = G_2$. The 14-dimensional Lie algebra of derivations $D : \mathbb{O} \rightarrow \mathbb{O}$ satisfying $D(xy) = D(x)y + xD(y)$ eliminates 35 of 49 degrees of freedom in the 8×8 A matrix, providing the tightest symmetry constraint of any division-algebra SSM.

D Implementation Details

All experiments use:

- Fixed-point arithmetic with 10^8 scaling factor (avoids floating-point non-determinism).
- Xoshiro128+ pseudorandom number generator with deterministic seeds.
- Box–Muller approximation via 12-sample CLT for normally distributed initialization.
- Taylor series ($n = 20$ terms) for $\exp$ and ($n = 10$ terms) for $\log$.
- Newton–Raphson (15 iterations) for $\sqrt{\cdot}$.
- Gradient clipping at ± 1.0 for hidden state gradients, ± 0.5 for A -matrix gradients.